Portfolio Assessment 3: Advanced ML - Regression/Classification using ML and Deep Learning

- Read the instructions carefully.
- Part A is common for everyone.
- Part B is based on the problem you have.
- Keep your datasets fixed for PART A and PART B. Don't change them midway, as you won't be able to make comparisons.

Select only one option to do, don't do all of them.

Select one option relevant to your dataset:	**Choose this**
Images (pixel data, medical images, handwritten digits, photographs) **OR**	**3a. Advanced ML Model and Deep Learning - CNN for Image Classification** **OR**
Sequential/Time-Series (sensor data, stock prices, text, audio, signal data) AND you have a **regression** target. **OR**	**3b. Advanced ML Model and Deep Learning - LSTM for Regression** **OR**
Sequential/Time-Series (sensor data, text, audio) AND you have a **classification** target. **OR**	**3c. Advanced ML Model and Deep Learning - LSTM for Classification** **OR**

Non-sequential data AND you have a **regression/classification** target **(Updated 27/04/2026)**	**3d. Advanced ML Model and Deep Learning - ANN for Classification/Regression**

Sample Code:

Some of the sample code has already been shared, and also relevant Kaggle Repos have been shared previously.

- **MLP_Regression.ipynb**
- **MLP_Classifcation.ipynb**
- **CNN_Image_Classifcation.ipynb**
- **DL_Models_ExampleCode.ipynb**
- **ANN_Regression_Classification.ipynb** **(Updated 27/04/2026)**

Submission Requirements

Format: One Jupyter notebook containing:

- Traditional ML model development and comparison.
- LSTM/CNN/ANN model development and evaluation.
- Saved model files: LSTM/CNN/ANN model (.h5) and traditional models (.pkl).
- GitHub repository with complete code.

Format: One PDF containing the explanations:

- Discuss Part A briefly with screenshots.
- Explain Part B in-depth with screenshots.
- Comparison visualisations (bar charts, loss curves, scatter plots).
- GenAI reflection in the format specified.

File Naming:

1. Portfolio_Assessment_3_[Studio_Number]_[YourName].pdf

Rubric

Portfolio Assessment 3 Rubric

Criteria	Ratings	Pts
Cover Page	**0 PtsSatisfactory**The cover page contains a title, task number, unit details, and submission date. **0 PtsUnsatisfactory**Cover page is missing information	0 pts
Acknowledgement of Country	**0 PtsSatisfactory**Statement is complete and appropriate **0 PtsUnsatisfactory**Statement is either incomplete or inappropriate	0 pts
Traditional ML Models	**10 to >7.5 PtsExcellent**Successfully implements all 2 models (Random Forest, MLP) with appropriate parameters; comprehensive evaluation with multiple metrics (Accuracy/F1/R^2/RMSE/MAE); professional comparison table; clear identification of best model **7.5 to >5.0 PtsProfiicient**Implements 2 models; basic evaluation; comparison table present; identifies best model **5 to >2.5 PtsDeveloping**Models not implemented correctly; minimal evaluation; unclear comparison. **2.5 to >0 PtsInsufficient**Missing models; no evaluation	10 pts
Traditional ML Visualization	**6 to >4.0 PtsExcellent**Professional, clear bar charts comparing primary metrics across all models; appropriate labels and titles; insightful annotations **4 to >2.0 PtsProfiicient**Acceptable visualizations; basic labels **2 to >0 PtsInsufficient**Unclear or poorly formatted visuals OR Missing visualizations	6 pts

Deep Learning - Data Preparation	**12 to >9.0 PtsExcellent**CNN: Correctly loads/preprocesses images; normalises pixel values; creates an appropriate train/validation split; documents class distribution and image dimensions. OR LSTM: Correctly creates sequences; appropriate sequence length justified; proper normalisation; clear documentation of data shape (samples, timesteps, features) OR MLP: Correctly prepares data; applies scaling; appropriate train/validation split; documents input shape **9 to >6.0 PtsProfiicient**Basic data preparation; some documentation; minor issues. **6 to >3.0 PtsDeveloping**Incomplete preparation; normalisation issues; poor documentation. **3 to >0 PtsInsufficient**No data preparation.	12 pts
Deep Learning - Architecture & Training	**16 to >12.0 PtsExcellent**CNN: Builds correct architecture (Conv2D, MaxPooling, Flatten, Dense, Dropout, Softmax); uses callbacks; saves model as .h5. OR LSTM: Builds correct architecture (2 LSTM layers with dropout, dense layers); uses callbacks; saves model as .h5. OR MLP: Appropriate architecture (multiple dense layers with dropout, proper input shape, matching output layer); uses callbacks; saves model **12 to >8.0 PtsProfiicient**Basic architecture; training completed; model saved. **8 to >4.0 PtsDeveloping**Incorrect architecture; training issues; model not saved. **4 to >0 PtsInsufficient**No deep learning model built.	16 pts

Deep Learning - Evaluation	**12 to >9.0 PtsExcellent**CNN: Test accuracy/loss; classification report; confusion matrix; training curves; sample predictions. OR LSTM (Regression): R^2, RMSE, MAE; loss curves; actual vs predicted scatter plot OR LSTM (Classification): Accuracy, F1-score; confusion matrix; training curves; compares with the best traditional ML model. OR MLP - Regression: R^2, RMSE, MAE; loss curves; actual vs predicted scatter plot. OR MLP - Classification: Accuracy, F1-score; confusion matrix; training curves. Compared with the best traditional ML model **9 to >6.0 PtsProfiicient**Basic evaluation, some plots, some comparison. **6 to >3.0 PtsDeveloping**Incomplete evaluation; missing plots; no comparison. **3 to >0 PtsInsufficient**No evaluation.	12 pts
Final Summary & Reflection	**8 to >6.0 PtsExcellent**Master results table comparing traditional ML and deep learning; clear identification of best overall model; thoughtful answers to all reflection questions. **6 to >4.0 PtsProfiicient**Results table present; basic reflections; identifies best model **4 to >2.0 PtsDeveloping**Incomplete table; superficial reflections. **2 to >0 PtsInsufficient**Missing summary	8 pts
GitHub Repository	**4 to >3.0 PtsExcellent**Models, results, and comments; GitHub History present. **3 to >1.0 PtsDeveloping**Models and results with fewer comments; some GitHub History. **1 to >0 PtsInsufficient**No repository or link was provided. GitHub commit all at once	4 pts

GenAI Reflection	**12 to >9.0 PtsExcellent**Excellent reflection with clear listing of ALL GenAI tools used; detailed description of specific tasks and prompts; clear code attribution; thoughtful learning reflection; acknowledges limitations and ethical considerations. **9 to >6.0 PtsProfiicient**Good reflection with most tools listed; describes tasks at high level; some code attribution; basic learning reflection. **6 to >3.0 PtsDeveloping**Vague GenAI mention; minimal task description; little attribution; superficial reflection **3 to >0 PtsInsufficient**No GenAI reflection or incomplete	12 pts
In person Studio Demo	**20 to >15.0 PtsExcellent**Exceptional demo. Able to answer questions related to problem definition, EDA-to-modelling journey, creative feature engineering, key findings, and/or professional presentation. **15 to >10.0 PtsProfiicient**Good demo. Was able to answer most topics with clarity. **10 to >5.0 PtsDeveloping**Basic demo; unable to answer key questions or unclear. **5 to >0 PtsInsufficient**Poor or no demo.	20 pts
Optional	**0 PtsSatisfactory**It can be used to give bonus marks based on the tutor's feedback. **0 PtsUnsatisfactory**Penalties can apply based on the tutor's feedback.	0 pts

Total points: 100